

Unit 2 — Exponents & Scientific Notation

★ Answer Key ★

Do not distribute to students until after the lesson.

Guided Notes — Fill-Ins

§1 Product & Quotient. Product: $x^3 \times x^4 = x^7$; $(2x^3)(3x^2) = 6x^5$; “Same base? ADD.” Quotient: $x^6 \div x^2 = x^4$; $(12x^7) \div (3x^2) = 4x^5$; “Same base? SUBTRACT.”

§2 Power. Power: $(x^2)^3 = x^6$; $(2x^3)^2 = 4x^6$; “Power to power? MULTIPLY.” Power of a Product: $(2x)^3 = 8x^3$; $(3ab)^2 = 9a^2b^2$; “Give EACH factor the power.” Power of a Quotient: $(2/3)^2 = \frac{4}{9}$; $(a/b)^3 = \frac{a^3}{b^3}$; “Power to top AND bottom.” Parentheses: $-3^2 = -9$ vs $(-3)^2 = 9$.

§3 Zero & Negative. Zero: $5^0 = 1$; $(3x^5)^0 = 1$; “ANY base⁰ = 1.” Negative: $2^{-3} = \frac{1}{8}$; $4x^{-3} = \frac{4}{x^3}$; “Flip to a fraction.” **Across the bar:** $\frac{x^{-2}}{y^{-4}} = \frac{y^4}{x^2}$.

Reciprocal: $(\frac{x^2}{y})^{-3} = (\frac{y}{x^2})^3 = \frac{y^3}{x^6}$.

2.2 Scientific Notation. Large $\rightarrow$ +exp; Small $\rightarrow$ -exp. Multiply: MULTIPLY coefficients, ADD exponents, adjust to 1-10. Divide: DIVIDE coefficients, SUBTRACT exponents. Examples: $(3 \times 10^4)(2 \times 10^5) = 6 \times 10^9$; $(8 \times 10^6) \div (4 \times 10^2) = 2 \times 10^4$; $(5 \times 10^3)(4 \times 10^2) = 20 \times 10^5 \rightarrow 2.0 \times 10^6$.

Practice 1 — Product & Quotient

1) x^{11} 2) $15x^8$ 3) $8a^5b^7$ 4) $-18x^5y^4$ 5) $30m^3n^6$ 6) x^5 7) $5a^5$ 8) $4x^3y^4$ 9) $4m^5n^2$ 10) $-6p^4q^3$ 11) $3x^5$ 12) $2a^4b^2$

Practice 2 — Power Rules

1) x^{20} 2) $16x^{12}$ 3) $-27a^6$ 4) x^6y^{15} 5) $16m^6n^4$ 6) $32x^{15}y^{20}$ 7) $\frac{x^8}{y^{12}}$ 8) $\frac{8a^3}{b^6}$ 9) $\frac{9x^8}{4y^2}$ 10) x^{24} 11) $72x^8y^7$ 12) $16a^{12}b^8$

Practice 3 — Zero & Negative

1) 1 2) 1 3) $\frac{1}{64}$ 4) $\frac{2}{x^5}$ 5) $\frac{y^2}{x^3}$ 6) $\frac{b^3c}{a^2}$ 7) $\frac{4n^5}{m^6}$ 8) $\frac{4y^3}{x^4}$ 9) $\frac{y^2}{x^2}$ 10) $\frac{27}{8}$ 11) $\frac{b^3}{a^6}$ 12) $\frac{y^4}{9x^2}$

Practice B — Conversions

1) 4.5×10^7 2) 3.2×10^{-4} 3) 602,000,000 4) 0.000091 5) 7.8×10^5 6) 6.7×10^{-6} 7) 34,500 8) 0.0012 9) 5.28×10^9 10) 0.025 11) 9.04×10^{-7} 12) 7,150,000 13) 6.34×10^{10} 14) 0.0000088

Practice C — Operations

1) 6×10^9 2) 2×10^4 3) 3.5×10^{10} 4) 3×10^3 5) 3.0×10^{10} 6) 6×10^4 7) 4×10^{11} 8) 2.0×10^4

Practice D — Word Problems

1) $(9.3 \times 10^7) \div (1.86 \times 10^5) = 5 \times 10^2$ s (500 s) 2) 40 $\times$ longer 3) $1 \times 10^5 = \$100,000$ 4) 5.2×10^{15} g 5) 3.7×10^8 m 6) 3×10^2 s (300 s) 7) $(2 \times 10^{12}) \div (4 \times 10^6) = 5 \times 10^5$ photos 8) $(3 \times 10^8)(6 \times 10^3) = 1.8 \times 10^{12}$ m

Cumulative Review

A • Exponent Rules: 1) x^{10} 2) $12x^7$ 3) $10a^4b^6$ 4) $4x^4$ 5) m^{15} 6) $16x^8y^{12}$ 7) $6a^3b^4$ 8) $\frac{x^{12}}{y^6}$

B • Conversions: 1) 7.2×10^6 2) 4.5×10^{-4} 3) 30,500,000 4) 0.000061 5) 9.1×10^4 6) 8×10^{-7} 7) 4,440,000 8) 0.0027

C • Operations: 1) 9×10^9 2) 4×10^6 3) 2×10^{10} 4) 6×10^6 5) 5.6×10^8 6) 4.8×10^2

D • Word Problems: 1) 4 $\times$ wider 2) 5×10^3 h (5,000 h) 3) 1×10^6 widgets 4) $(6 \times 10^{-5}) \div (3 \times 10^{-7}) = 2 \times 10^2 = 200 \times$ larger

E • Negative Exponents: 1) $\frac{1}{25}$ 2) $\frac{3}{x^4}$ 3) $\frac{y^3}{x^5}$ 4) $\frac{b^3}{a^3}$ 5) $\frac{5b^6}{a^5}$ 6) $\frac{y^2}{4x^4}$